SHOGUN-SYNC

Total War: Shogun 2 Multiplayer Synchronization Tool

System Errors & Troubleshooting

Diagnostic Codes & Recovery Manual

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Module: Diagnostics (`errors.pdf`)

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1 Diagnostic Architecture

Shogun-Sync employs a unified, 3-digit error classification scheme (`Error 001` through `Error 020`) across all C modules. When an error occurs, the program outputs a message formatted as:

```
Error <XXX>:  <MESSAGE>
```

2 System Error Codes Matrix

Code	Identifier	Description
001	`SAVE_NOT_FOUND`	The target `.save_multiplayer` file was not found in the save directory.
002	`FILE_WRITE_FAILED`	Unable to open or write a file to disk.
003	`COMPRESSION_FAILED`	Zlib stream compression operation returned an error.
004	`COMPRESSION_INIT_FAILED`	Zlib `deflateInit` failed to initialize the compression stream.
005	`APPDATA_NOT_FOUND`	The `%APPDATA%` environment variable is not defined in the operating system.
006	`LOCALAPPDATA_NOT_FOUND`	The `%LOCALAPPDATA%` environment variable is not defined.
007	`INIT_FAILED`	Could not create the hidden `.sync` metadata directories on disk.
008	`SCAN_FAILED`	Failed to open or read the multiplayer save directory.
009	`CTRL_HANDLER_FAILED`	Win32 console control handler failed to attach.
010	`DECOMPRESS_INIT_FAILED`	Zlib `inflateInit` failed to initialize the decompression stream.
011	`DECOMPRESS_STREAM_FAILED`	Zlib decompression error encountered during stream inflation.
012	`DECOMPRESS_WRITE_FAILED`	Failed to write restored uncompressed save data to the destination path.
013	`DECOMPRESS_OPEN_FAILED`	Could not open a compressed blob file for decompression.
014	`HOSTNAME_FAILED`	Windows Sockets failed to resolve the local machine identity.
015	`CONN_FAILED`	WebSocket connection to the relay server failed or timed out.
016	`INVALID_PACKET`	Received malformed packet or incomplete Auto-Resync binary payload.
017	`MEMORY_ALLOC_FAILED`	Heap memory allocation (`malloc`) failed.
018	`COMMIT_FAILED`	Failed to write commit manifest or update reference pointer.
019	`RESYNC_RESTORE_FAILED`	Failed to decompress a received save payload during Auto-Resync.
020	`INVALID_ARGUMENTS`	Required command-line arguments are missing or malformed.

3 Subsystem Classification

3.1 Filesystem & Saves (001, 005, 006, 007, 008, 009, 018)

Errors related to finding, opening, reading, and writing game save files and metadata in:

`%APPDATA%/TheCreativeAssembly/Shogun2/`

- **Error 001:** Triggered when attempting a manual sync before any multiplayer campaign save exists.
- **Error 007:** Triggered if permission is denied when creating the hidden `.sync` repository folder.
- **Error 018:** Failed to record the commit tree manifest or update `refs/latest`.

3.2 Compression & Memory (002, 003, 004, 010, 011, 012, 013, 017)

Errors within the Zlib integration layer and memory subsystem:

- **Errors 003 & 004:** Occur during deflate operations when packing saves into objects.
- **Errors 010, 011, 012, 013:** Occur during inflate operations when restoring a save.
- **Error 017:** Indicates insufficient free RAM to hold the save buffer.

3.3 Network & Lobby (014, 015, 016, 018, 019, 020)

Handles multiplayer room creation, joining, and packet exchange:

- **Error 015:** Occurs when the program cannot reach the server (`wss://shogun2-sync.duckdns.org`) due to internet issues, an invalid room code, or a wrong password.
- **Error 016:** Received a malformed binary frame that does not contain the required 64-byte filename header or contains invalid path characters.
- **Error 019:** The client received a binary save payload from the host, but decompression failed.
- **Error 020:** The user ran `create` or `join` without providing the necessary command line parameters.